MILITARY SPECIFICATION

TEST STAND, OXYGEN SYSTEM COMPONENTS

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of an oxygen components test stand used for testing and evaluating aircrewmen's oxygen breathing regulators and oxygen system components.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-N-411	- Nitrogen, Technical.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-C-843	- Cushioning, Material, Cellulosic.
PPP-T-60	- Tape, Packaging, Waterproof.

MILITARY

MIL-C-104	- Crate, Wood, Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-I-6866	- Inspection, Penetrant Method of.
MIL-I-6868	- Inspection Process, Magnetic Particle.
MIL-P-7105	- Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements for.
MIL-S-7742	- Screw Threads, Standard, Optimum Selected Series, General Specification for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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MIL-S-8512 - Support Equipment, Aeronautical, Special General Specification for the Design of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-143 - Standards and Specifications, Order of Precedence for the Selection of.
MIL-STD-278 - Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping and Pressure Vessels in Ships of USN.
MIL-STD-831 - Test Reports, Preparation of.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

1316AS100 - Oxygen System Components Test Stand.

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND

NAVSEA 0900-LP-003-800 - Surface Inspection Acceptance Standards for Metals.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the acquiring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committe, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL BUREAU OF STANDARDS (NBS)

Report No. 4696 - Oxygen Regulator Test Stand.

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(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.4).

3.2 Oxygen system components test stand. The requirements for the oxygen system components test stand shall be in accordance with MIL-S-8512 and as specified herein.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials. Materials shall conform to applicable specifications and shall be as specified herein and on applicable drawings. Materials which are not covered by specifications or which are not specifically described herein shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.4.1 Metal parts. All metal parts shall be of a corrosion resistant material or treated in a manner to render them adequately resistant to corrosion.

3.4.1.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.2 Nonmetallic materials. Any nonmetallic material that is adversely affected by continued use with oxygen shall not be used.

3.4.2.1 Age. Elastomer components, except silicone, shall be not more than 12 months old from the cure date of manufacture to the date of delivery to any Government Service.

3.4.2.2 Fungus-proof materials. Materials which are not nutrients for fungi shall be used to the greatest extent practicable. In cases where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent.

3.4.3 Protective treatment. When materials are used in the construction of the test stand that are subject to deterioration when exposed to environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective coating which might crack, chip, or scale during normal service life or under extremes of environmental conditions shall not be used.

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3.5 Standard parts. MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the manufacturer's drawings by their part numbers. Parts which are not covered by specifications or military standards, or which are not specifically described herein or on the drawings, shall be completely described on manufacturer's drawings.

3.6 Design. The oxygen system component test stand shall provide a high and low pneumatic pressure and a vacuum system with the necessary valving and instrumentation to measure, test and evaluate the performance and operating characteristics of oxygen system components from sea level up to a simulated altitude of 150,000 feet. The pneumatic systems shall operate from the common external nitrogen pressure source of up to 3,000 psi. The vacuum system shall include an altitude chamber and a vacuum pump. The test stand shall be in accordance with the drawings listed on Drawing 1316AS100 and as specified herein.

3.7 Construction. The test stand shall be so constructed that parts will not work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, and service. Welding, riveting, or other methods of assembly may be used, providing subassemblies can be removed as required for service, maintenance, overhaul, inspection or replacement.

3.7.1 Locking devices. Screws, pins, bolts, etc., shall be furnished with standard locking devices where required. Safety wire, self-locking nuts, cotterpins and lockwashers shall be acceptable.

3.7.2 Threads. All straight screw threads shall conform to the requirements of MIL-S-7742. Pipe threads, where permitted herein, shall conform to MIL-P-7105.

3.7.3 Welding. Welding shall be performed in accordance with MIL-STD-278, class M-2. The welds shall be inspected and accepted in accordance with MIL-I-6868, MIL-I-6866 and NAVSEA 0900-LP-003-800.

3.8 Performance.

3.8.1 Leakage tests.

3.8.1.1 High pressure section (phase 1). The high pressure section, when tested as specified in 4.6.2.1, shall not show any evidence of leakage and there shall not be any pressure reading on the regulated high and low pressure gages.

3.8.1.2 High pressure section (phase 2). The high pressure section, when tested as specified in 4.6.2.2.1 through 4.6.2.2.3, shall not show any evidence of leakage and the regulated low pressure gage shall not exceed 175 ! 5 psi.

3.8.1.3 High pressure section (phase 3). The high pressure section, when tested as specified in 4.6.2.3, shall not show any evidence of leakage.

3.8.1.4 Low pressure system.

3.8.1.4.1 Pressure decay. The low pressure system, when tested as specified in 4.6.2.4.1, shall maintain the 70 psi indicated on the regulated low pressure gage.

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3.8.1.4.2 Inlet pressure "on/off" valve. The inlet pressure "on/off" valve, when tested as specified in 4.6.2.4.2, shall not show any flow indication on the high range leakage rotameter.

3.8.1.4.3 Leakage control valve. The leakage control valve, when tested as specified in 4.6.2.4.3, shall not show any flow indication on low range leakage rotameter.

3.8.1.4.4 Gage protector. The nitrogen input pressure guage shall indicate a pressure of 145 ± 5 psi when the gage protector is tested as specified in 4.6.2.4.4.

3.8.2 Electrical ground continuity test. The test stand, when tested as specified in 4.6.3, shall have an ohm reading of zero ohms.

3.8.3 Motor rotation test. The motor, when tested as specified in 4.6.4, shall rotate in a clockwise direction when facing pulley end of motor.

3.8.4 Leakage test (altitude chamber).

3.8.4.1 Altimeters. The altimeters, when tested as specified in 4.6.5.1, shall indicate the same altitude.

3.8.4.2 Altitude chamber. The time required for the chamber altitude to drop from 52,000 to 51,000 feet, when tested as specified in 4.6.5.2, shall be not less than 20 minutes.

3.9 Calibration graphs. A calibration graph shall be prepared for an output flow calibration curve, input flow calibration curve, vent flow calibration curve and orifice calibration curve. Each curve shall be on a separate graph and approximately identified. The calibration procedure 3.9.1 shall be used to acquire the data for plotting the curves.

3.9.1 Flowmeter calibration. Prior to starting calibration, prepare a flowmeter calibration test sheet (figure 1). Using the secondary master graph, determine the pressure in inches H_2O for all corrected flows for each altitude except ground level and enter the amount in the secondary master column next to the applicable corrected flow. The calibration procedure shall be performed in the following steps:

a. Close all test stand "on/off" valves and open the nitrogen supply source.
b. Set pressure selector valve (D, figure 2) to "Hg" position.
c. Set flow selector valve (M) to regulator position.
d. Set reference pressure selector valve (O) to altitude chamber position.
e. Connect suction hose of secondary master flow element (12) to piezometer (26) at output connection (23) in chamber. (See figure 3.)
f. Start vacuum pump.
g. Adjust output valve (C) to obtain required flow indicated in inches of water (equivalent to 25 lpm) at ground level on the secondary master flow element (12). Record in inches on the calibration sheet the inches of water indicated on the output flowmanometer (1) in the output manometer column (figure 1) for 25 lpm at ground level. Repeat the above procedure using the indicated pressure at each of the ground level floors.

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h. Set the output valve (C) to the "off" position and disconnect secondary master flow element (12).
i. Insert the calibration orifice, into piezometer (26) at the output connection (23) after verifying that the serial number of orifice coincides with the test stand serial number.
j. Set the pressure selector valve (D) to "H_2O" position.
k. Adjust output valve (C) to obtain inches of water reading on output flow manometer (1) for ground level only, as recorded in step (g). Read and record inches of water indicated on pressure suction manometer (4), on test sheet under column title "orifice".
l. Close output valve (C).
m. Set the pressure selector valve (D) to "Hg" position.
n. Connect absolute test manometer (17, figure 4), to 200 cc leakage connection (20) at rear of test stand and the secondary "master flow element" to "flowmeter input calibration port" on front of test stand.
o. Remove orifice and chamber input plug at input connection (22, figure 2).
p. Close chamber door.
q. Adjust vacuum control valve (B) until an altitude of 20,000 feet is indicated on the altimeter low range indicator (13), and then close vacuum control valve (B).
r. Adjust ouput valve (C) to obtain required inches of water on output flowmeter (1) as recorded in step (g). Adjust input valve (A) to stabilize at 20,000 feet.
s. Record the reading indicated on output and input flow manometers (1 and 2) on calibration sheet under column headings "output and input."
t. Continue recording inches of water indications on output and input flow manometers (1 and 2) for all flows and altitudes indicated on calibration sheet.
u. Shut off output valve (C) and input valve (A) simultaneously.
v. Shut off vacuum pump.
w. Open chamber bleed valve (K) and return chamber simulated altitude to ground level.
x. Insert the calibration orifice into the piezometer (26) at output connection (23).
y. Set pressure selector (D) to "H_2O" position.
z. Ascertain that chamber bleed valve (K) is closed and start vacuum pump.
aa. Repeat steps (q) through (s), except use inches of water from "manometer in. water output" column for equivalent flows at 20,000 feet.
bb. Observe and record readings indicated on pressure suction manometer (4) for 20,000 feet under column title "orifice".
cc. Secure from test.

3.10 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

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3.11 Workmanship. The unit shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to cleanliness, freedom from defects, burrs, sharp edges, accuracy of dimensions, radii of fillets, machine screw assemblies, welding, brazing and riveting, neatness and thoroughness of soldering, wiring, plating and marking of parts and assemblies.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection. First article inspection consists of examinations and tests performed on samples which are representative of the production item after award of a contract to determine that the production item conforms to the requirements of this specification (see 3.1 and 4.3).
b. Quality conformance inspection. Quality conformance inspection consists of examinations and tests performed on individual products or lots to determine conformance of the products or lots to the requirements set forth in this specification (see 4.4).

4.3 First article inspection. The first article inspection of the test stand shall consist of all the examinations and tests of this specification.

4.3.1 First article sample. The first article sample shall consist of one complete test stand representative of construction, workmanship, components and materials to be used during production. The contractor shall give the acquiring activity a minimum of 30 days advance notice as to when the first article tests are to be conducted.

4.3.2 First article inspection report. Following completion of the first article inspection, a test report which includes all tests and examinations as specified herein shall be prepared in accordance with MIL-STD-831 and shall be furnished to the acquiring activity as specified in the contract or purchase order (see 6.2).

4.4 Quality conformance inspection. Quality conformance inspection shall be performed on each test stand. The quality conformance testing shall be successfully completed before being accepted by the Government. The sample size, acceptance criteria, tests and examinations required for the test stand shall be as specified in table I.

4.5 Test conditions.

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4.5.1 Temperature and pressure. Unless otherwise specified, tests shall be conducted at local ambient temperature and barometric pressure. The temperature and barometric pressure shall be recorded at the time of inspection. This information shall be available for computation of test data, where required, to normal temper-- ature and pressure (NTP) conditions. NTP conditions are 70°F (21°C) and 29.92 inches of mercury (101.3kPa). Test instruments shall be calibrated or adjusted according to their required usage in conducting individual tests.

4.5.2 Gas. Unless otherwise specified, the gas used in testing shall conform to nitrogen, BB-N-411, type I, class I, grade B.

4.5.3 Power. Unless otherwise specified, the test stand shall be tested with a 208 - 220 volt, 3 phase 60 ! 2 Hertz AC power source.

4.5.4 Test equipment for calibration procedure.

4.5.4.1 Secondary master flow element. The secondary master flow element shall consist of a 0 to 23 inch manometer, Model 411-23, F. W. Dwyer Manufacturing Company, or equivalent, connected to a 0 to 300 lpm flow element Model 10-10-5, Calibrating and Measuring Equipment, Inc., or equivalent with an accuracy of 1/2 percent.

4.5.4.2 Absolute test manometer. The absolute test manometer shall be a Model 411-23, F. W. Dwyer Manufacturing Company, or equivalent.

4.6 Inspection methods.

4.6.1 Visual examination.

4.6.1.1 Test stand. Every test stand shall be examined visually for critical defects to determine conformance to this specification. Every test stand, selected as a sample unit from the lot, shall be visually examined for minor defects to determine conformance to this specification. The classification of defects, table II, shall be used to classify the defects found.

4.6.1.1.1 Dimensions. The test stand shall be checked dimensionally to determine conformance to the dimensions specified on the applicable drawing.

4.6.2 Leakage tests. Connect a nitrogen gas supply, 1,800 to 2,200 psi pressure to the test stand nitrogen input connector (15, figure 2). Position all the test stand valves in the closed position. The leakage tests shall then be performed in the following sequence with each control adjusted only as specified.

4.6.2.1 High pressure section (phase 1).

4.6.2.1.1 Step one. Slowly open the nitrogen gas supply valve until the test stand supply pressure gage (9) reads 500 ! 25 psi. Close nitrogen gas supply valve and apply leak detection compound to all connections in the test stand high pressure section (see figure 2) and check for leaks. The test stand shall pass the requirements specified in 3.8.1.1.

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4.6.2.1.2 Step two. Repeat 4.6.2.1.1, except increase the test stand supply pressure gage reading to 1,000 ! 50 psi. The test stand shall pass the requirements specified in 3.8.1.1.

4.6.2.1.3 Step three. Repeat 4.6.2.1.1, except increase the test stand supply pressure gage reading to 1,800 to 2,200 psi. The test stand shall pass the requirements specified in 3.8.1.1.

4.6.2.2 High pressure section (phase 2).

4.6.2.2.1 Step one. Set the leakage selector valve (F) to the "high" position and open the nitrogen gas supply valve. Slowly adjust the test stand high pressure regulator (Q) until the regulated high pressure gage (10) reads 500 ± 25 psi. Record the regulated low pressure gage (11) reading and then apply leak detection compound to connections in high pressure section (system bleed port) and check for leaks. The system and the gage protector shall pass the requirements specified in 3.8.1.2.

4.6.2.2.2 Step two. If no leaks are detected during 4.6.2.2.1, continue to adjust the high pressure regulator (Q) until the regulated high pressure gage (10) reads 1,500 psi and repeat the application of leak detection compound, as in 4.6.2.2.1, and check for leaks. The system shall pass the requirements specified in 3.8.1.2.

4.6.2.2.3 Step three. If no leaks are detected during 4.6.2.2.2, continue to adjust the high pressure regulator (Q) until the regulated high pressure gage (10) reads 1,800 to 2,200 psi and repeat the application of leak detection compound, as in 4.6.2.2.1, and check for leaks. The system shall pass the requirements specified in 3.8.1.2.

4.6.2.3 High pressure section (phase 3).

4.6.2.3.1 Step one. With the high pressure regulator (Q) adjusted to 1,800 to 2,200 psi, turn the flow selector valve (M) to the regulator position and open the vent ambient valve (I). Attach hose to inlet of the vent ambient valve (I), immerse the free end of the hose in water for a minimum period of one minute, and then check for leakage. The system shall pass the requirements specified in 3.8.1.3.

4.6.2.3.2 Step two. Close vent ambient valve (I) and cap nitrogen input connection (18) in the altitude chamber. After one minute, record the nitrogen input gage (22) reading in the altitude chamber. The system shall pass the requirements specified in 3.8.1.3.

4.6.2.3.3 Step three. Attach a hose to the low pressure connection (19) in the altitude chamber. Immerse the free end of the hose in water for a minimum period of one minute and check for leakage. The system shall pass the requirements specified in 3.8.1.3. Adjust high pressure regulator (Q) to zero pressure and bleed system using system bleed valve (S) until the regulated high and low pressure low pressure gages (10 and 11) indicate zero pressure. Close all test stand valves.

4.6.2.4 Low pressure system.

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4.6.2.4.1 Pressure decay. Adjust low pressure regulator (N) until the regulated low pressure gage (11) indicates 70 psi. Open inlet pressure "on/off" valve (L), rotate low pressure regulator (N) to zero pressure position and after a minimum of 5 minutes record the pressure reading on the regulated low pressure gage (11). The low pressure system shall pass the requirements specified in 3.8.1.4.1.

4.6.2.4.2 Inlet pressure "on/off" valve. Adjust low pressure regulator (N) until the regulated low pressure gage (11) indicates 70 psi, if necessary. Close inlet pressure valve (L), open leakage "on/off" valve (G) and observe high range leakage rotameter (8) for a flow indication. The inlet pressure "on/off" valve shall pass the requirements specified in 3.8.1.4.2.

4.6.2.4.3 Leakage control valve. Turn leakage control valve (F) to "low" position and observe low range leakage rotameter (7) for a flow indication. The leakage control valve shall pass the requirements specified in 3.8.1.4.3.

4.6.2.4.4 Gage protector. Turn leakage "on/off" valve (G) to "off" position and open inlet pressure "on/off" valve (L). Adjust the low pressure regulator (N) until the regulated low pressure gage (11) indicates 160 ! 5 psi and then record the input pressure gage (27) reading. The gage protector shall pass the requirements specified in 3.8.1.4.4. Adjust the low pressure regulator (N) to the zero pressure position and remove cap from the nitrogen input connection (18) in the altitude chamber.

4.6.3 Electrical ground continuity test. Connect one lead from a volt-ohm meter, ohm scale accuracy of ! 3 percent at full scale, to the ground pin of the test stand electical power connector. The other lead from the volt-ohm meter shall be connected to the bare metal of the test stand valve bodies. The test stand shall pass the requirements specified in 3.8.2.

4.6.4 Motor rotation test. Remove belt guard, loosen the four mounting bolts on the vacuum pump motor and slide the motor forward until the belt drive between the motor and vacuum pump is loose. Remove the belts. Energize the motor and record the direction of rotation of the motor pulley. The motor shall pass the requirements specified in 3.8.3. Replace the belts on the vacuum pump and motor pulleys, slide the motor back until the correct belt tension is achieved and then tighten the four bolts on the motor.

4.6.5 Leakage test (altitude chamber).

4.6.5.1 Altimeters. Connect a nitrogen gas supply, 1,800 to 2,200 psi pressure, to the test stand nitrogen input connector (15). Close all "on/off" valves and open the nitrogen gas supply valve. The test stand supply gage (9) shall indicate a pressure of 1,800 to 2,200 psi. Place the pressure equalizer valve (Z) in the "off" position, the pressure selector valve (D) in the "Hg" position, and the reference pressure selector valve (O) in the altitude chamber position. Open the flutter dampener (J) and suit simulator reference shut off valve (R) one turn. Remove the air inlet plug from the input port (22) in the altitude chamber, close the altitude chamber door and start the vacuum pump. Adjust the vacuum control valve (B) and the chamber bleed valve (K) until an altitude of 30,000 feet is indicated on the altimeter low range indicator (13). Record the altitude indicated on the altimeter high range indicator (12). The test stand shall pass the requirements specified in 3.8.4.1.

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4.6.5.2 Altitude chamber. Close chamber bleed valve (K), adjust vacuum control valve (B) until an altitude of 52,000 feet is indicated on the altimeter high range indicator (12) and then close the vacuum control valve (B). After a minimum of two minutes stabilization, record the time required for the chamber altitude to drop from 52,000 feet to 51,000 feet. The test stand shall pass the requirements specified in 3.8.4.2. Turn off the vacuum pump by depressing pump control switch, open chamber bleed valve (K) and altitude chamber door, and then close vacuum control valve (B).

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Each test stand shall be preserved and packaged in accordance with method IIA of MIL-P-116. Test stand accessories shall be packaged individually in domestic fiberboard containers conforming to PPP-B-636. Adequate cushioning material conforming to PPP-C-843 shall be provided to prevent movement or damage. All openings of the test stand and pneumatic system shall be plugged, capped, or covered with barrier material conforming to MIL-B-121, grade A. When MIL-B-121 barrier material is used, it shall be secured in place with tape conforming to PPP-T-60.

5.1.2 Level C. Each test stand shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A and B. Each test stand and accessories preserved and packaged as specified in 5.1.1 shall be packed in a container conforming to MIL-C-104, type I, class 2. Containers shall be of uniform size and shape and of minimum tare and cube. Closure and strapping shall be in accordance with MIL-C-104.

5.2.2 Level C. Each test stand and accessories preserved and packaged as specified in 5.1.2 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall conform to Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Physical protection. Interior cushioning, anchoring, blocking and bracing shall be in accordance with MIL-STD-1186. Tests: Edgewise-drop test or cornerwise drop test shall be in accordance with Appendix A of the specification. Tests are not required when level C packing is specified.

5.4 Marking. In addition to any special marking required by the contract or order, interior packages and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The test stand covered by this specification is intended for testing aviator's oxygen breathing regulators and oxygen system components.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Whether first article inspection is required and where the samples should be delivered (see 4.3.1).
c. Level of preservation, packaging and packing (see 5.1 and 5.2).
d. Items of data required (see 6.3).

6.3 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.3.2	First Article Inspection Reports	DI-T5329, Inspection Test Report

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the acquiring activity or as directed by the contracting officer).

6.4 First article. When a first article is required, it shall be tested under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The first article should be a first production item. The contracting officer should include specific instructions in all acquisition instruments, regarding arrangements for examinations, test and approval of the first article.

Preparing Activity;
Navy - AS

(Project No. 4920-N375)

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TABLE I. Sample size, acceptance criteria, tests and examination of the test stand.

Inspection	Method	Sample size	Acceptance criteria
Visual examination (see classification of defects)	4.6.1	Every test stand for critical defects. Inspection level II for minor defects. 1/	Reject all units with any critical defect. An acceptable qualtiy level of 2.5 defects per hundred units for minor defects.
Dimensions	4.6.1.1.1	Inspection level S-2 1/	Acceptance number zero, rejection number 1.
Leakage tests	4.6.2	Every test stand	Reject all defective units. 2/
Electrical ground continuity test	4.6.3	Every test stand	Reject all defective units. 2/
Motor rotation test	4.6.4	Every test stand	Reject all defective units. 2/
Leakage (altitude chamber)	4.6.5	Every test stand	Reject all defective units. 2/

1/ The sample size shall be based on the applicable size code letter corresponding to the specified inspection level of MIL-STD-105.

2/ A failure report shall be generated for all rejected units and included with the final report. Corrective action shall be taken and the units retested for the applicable test.

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TABLE II. Classification of defects for visual examination of the test stand.

Critical	Minor
1. Material imperfections - foreign matter embedded.	201. Marking - missing, insufficient, incorrect or illegible.
2. Surface - unclean, rough, misaligned or contains cracks, nicks or other flaws.	202. Color not as specified.
3. Any component missing, malformed, fractured or otherwise damaged.	
4. Any component loose or otherwise not securely retained.	
5. Incorrect assembling or improper positioning of components.	
6. Any functioning part that works with difficulty.	
7. Faulty workmanship or other irregularities.	

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FLOWMETER CALIBRATION USING AIR CURVE

Date _____
S/N _____
Insp. _____

ALTITUDE (X 1000 FT)	FLOW RATE LPM AT 14.7 PSIA		SECONDARY MASTER(12) In. H_2O	MANOMETER (1) In. H_2O OUTPUT	MANOMETER (2) In. H_2O INPUT	MANOMETER (4) In. H_2O ORIFICE
	STANDARD	CORRECTION FOR ALTITUDE				
GROUND	25					
	50					
	75					
	100					
	125					
20	25	11.483				
	50	22.967				
	75	34.451				
	100	45.934				
	125	57.418				
30	25	7.420				
	50	14.841				
	75	22.261				
	100	29.682				
	125	37.102				
40	25	4.627				
	50	9.255				
	75	13.883				
	100	18.511				
	125	23.139				
50	25	2.871				
	50	5.743				
	75	8.614				
	100	11.486				
	125	14.357				

Input Flow Element S/N _____

Output Flow Element S/N _____

Vent Flow Element S/N _____

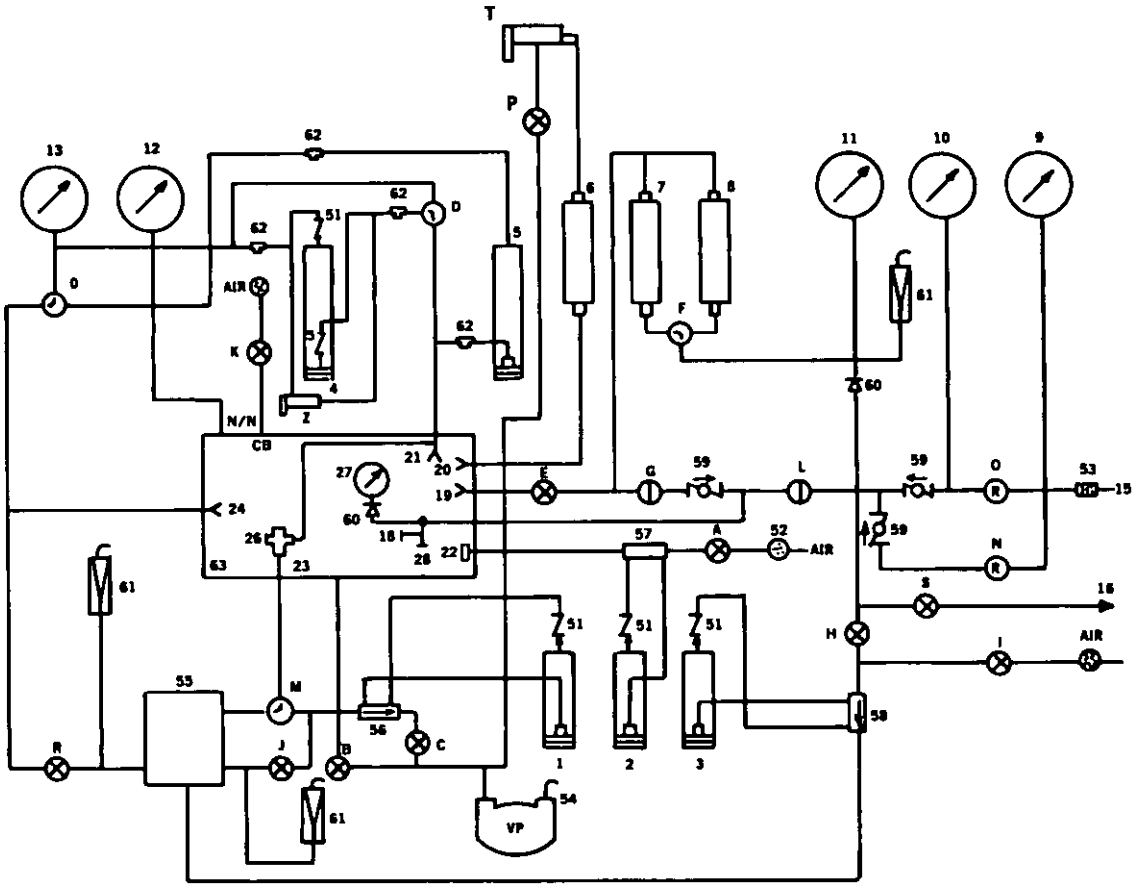
Pump S/N _____

NOTE:

Corrected flows for altitude were derived from National Bureau of Standards Report No. 4696, page 60, Section 5-1-3.

FIGURE 1. Calibration sheet.

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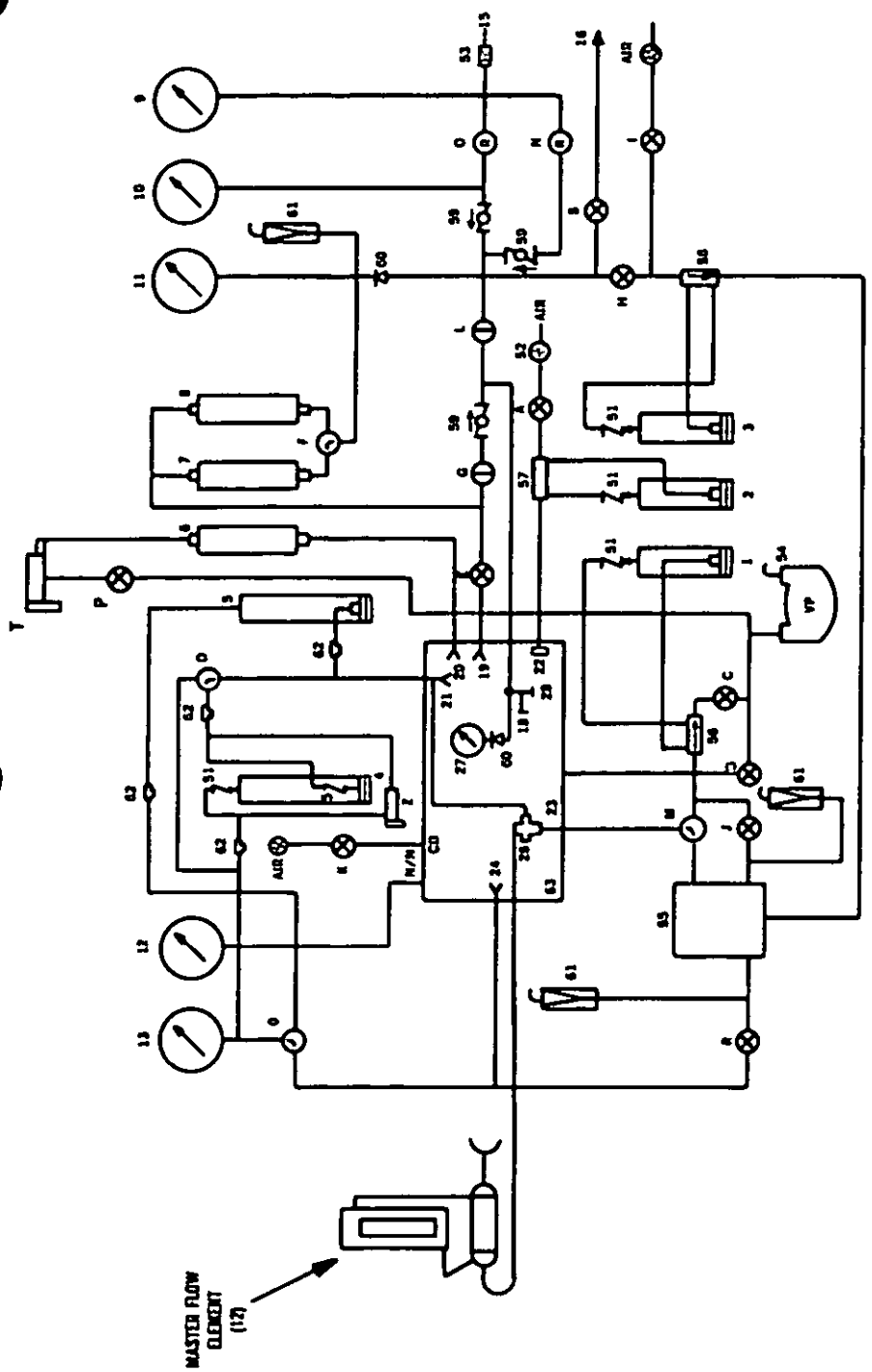
A. INPUT
B. VACUUM CONTROL
C. OUTPUT
D. PRESS. SELECTOR
E. LEAKAGE CONTROL
F. LEAKAGE CONTROL
G. LEAKAGE ON/OFF
H. VENT PRESS.
I. VENT AMBIENT
J. DAMPENER
K. CHAMBER BLEED
L. INLET PRESS. ON/OFF
M. FLOW SELECTOR
N. LOW PRESS. REGULATOR
O. REF. PRESS. SELECTOR
P. INWARD LEAKAGE VALVE
Q. HIGH PRESS. REGULATOR
R. SUIT SIMULATOR REF. SHUT-OFF
S. SYSTEM BLEED
T. OVERBOARD ON/OFF VALVE
Z. PRESS. EQUALIZER
MN. CHAMBER REF. PORT
VP. VACUUM PUMP

1. OUTPUT FLOW
2. INPUT FLOW
3. VENT FLOW
4. PRESS. SUCTION
5. HG. MANOMETER
6. OVERBOARD LEAKAGE ROTAMETER
7. LOW RANGE LEAKAGE ROTAMETER
8. HIGH RANGE LEAKAGE ROTAMETER
9. SUPPLY PRESS. (0-3000 PSI)
10. REGULATED HIGH PRESS.
11. REGULATED LOW PRESS.
12. ALTM HIGH RANGE
13. ALTM LOW RANGE
16. N_2 INPUT CONNECTOR
16. SYSTEM BLEED PORT
18. N_2 INPUT CONNECTION
19. LOW PRESS. CONNECTION
20. 200 CC LEAKAGE CONNECTION
21. REF. TAP CONNECTION
22. INPUT PORT

23. OUTPUT
24. HELMET REF TAP
26. PIEZOMETER
27. N_2 INPUT PRESS GAGE
28. TEE CONNECTOR
51. CHECK VALVE (FLOW)
52. SCREEN FILTER
53. MICRON FILTER
54. VACUUM PUMP VENT
55. SUIT SIMULATOR
56. FLOW ELEMENT
57. FLOW ELEMENT
58. FLOW ELEMENT
59. CHECK VALVE
60. GAGE PROTECTOR
61. RELIEF VALVE
62. LINE TRAP
63. ALTITUDE CHAMBER

FIGURE 2. Leakage test set up.

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A. INPUT
B. VACUUM CONTROL
C. OUTPUT
D. PRESS. SELECTOR
E. LEAKAGE CONTROL
F. LEAKAGE CONTROL
G. LEAKAGE ON/OFF
H. VENT PRESS.
I. VENT AMBIENT
J. DAMPENER
K. CHAMBER BLEED
L. INLET PRESS. ON/OFF
M. FLOW SELECTOR
N. LOW PRESS. REGULATOR
O. REF. PRESS. SELECTOR
P. INWARD LEAKAGE VALVE
Q. HIGH PRESS. REGULATOR
R. SUIT SIMULATOR REF. SHUT-OFF
S. SYSTEM BLEED
T. OVERBOARD ON/OFF VALVE
Z. PRESS. EQUALIZER
MN. CHAMBER REF. PORT
VP. VACUUM PUMP

1. OUTPUT FLOW
2. INPUT FLOW
3. VENT FLOW
4. PRESS. SUCTION
5. HG. MANOMETER
6. OVERBOARD LEAKAGE ROTAMETER
7. LOW RANGE LEAKAGE ROTAMETER
8. HIGH RANGE LEAKAGE ROTAMETER
9. SUPPLY PRESS. (0-3000 PSI)
10. REGULATED HIGH PRESS.
11. REGULATED LOW PRESS.
12. ALTM HIGH RANGE
13. ALTM LOW RANGE
16. N_2 INPUT CONNECTOR
18. SYSTEM BLEED PORT
19. N_2 INPUT CONNECTION
20. LOW PRESS. CONNECTION
20. 200 CC LEAKAGE CONNECTION
21. REF. TAP CONNECTION
22. INPUT PORT
23. OUTPUT
24. HELMET REF TAP
26. PIEZOMETER
27. N_2 INPUT PRESS GAGE
28. TEE CONNECTOR
51. CHECK VALVE (FLOW)
52. SCREEN FILTER
53. MICRON FILTER
54. VACUUM PUMP VENT
55. SUIT SIMULATOR
56. FLOW ELEMENT
57. FLOW ELEMENT
58. FLOW ELEMENT
59. CHECK VALVE
60. GAGE PROTECTOR
61. RELIEF VALVE
62. LINE TRAP
63. ALTITUDE CHAMBER

FIGURE 3. Output flowmeter and orifice calibration test set up (ground only).

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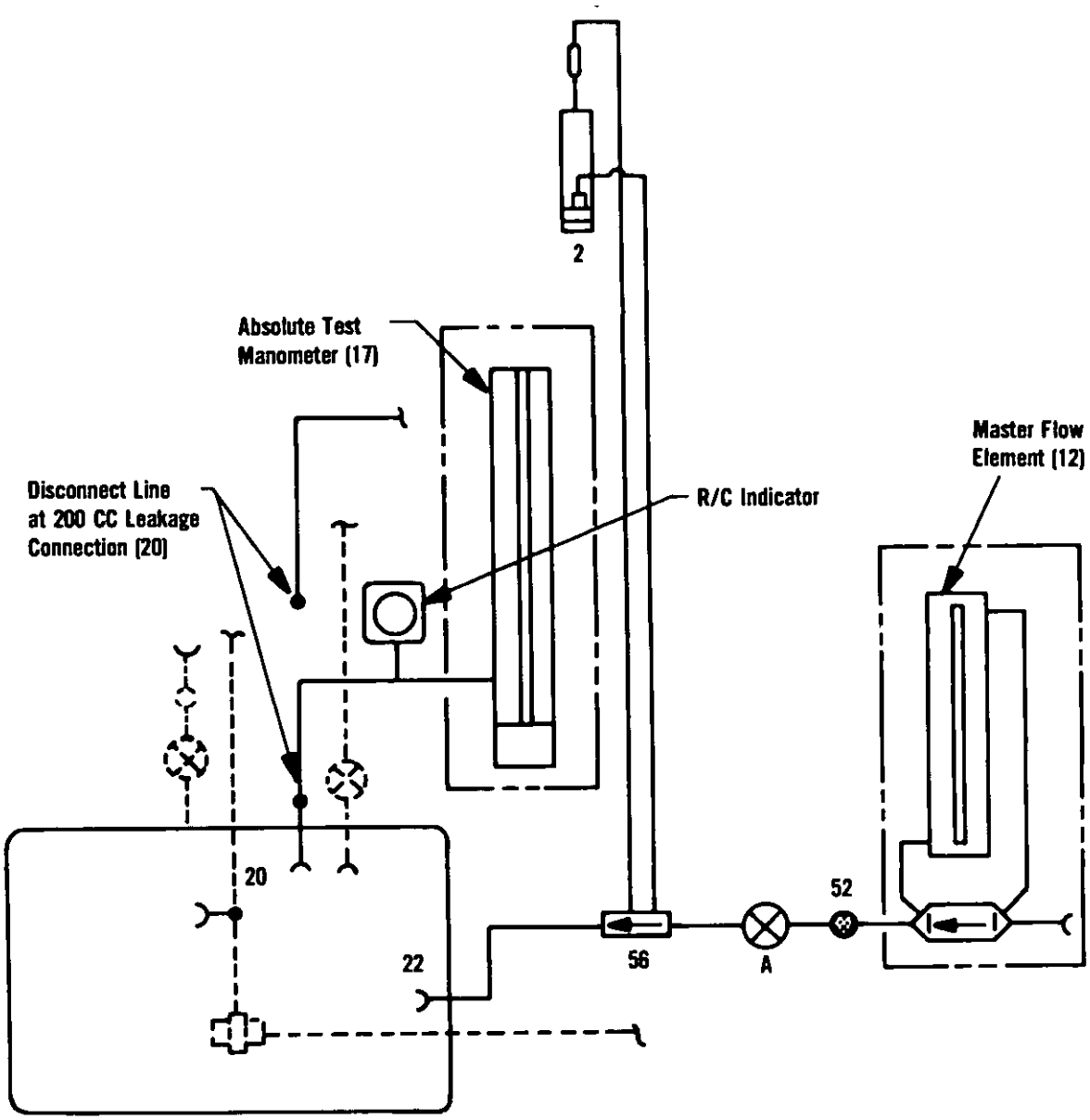


FIGURE 4. Input and output flowmeter and orifice calibration test set up altitude.

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